Design and Simulation of a Buck + LDO Power Architecture

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1. Objective

This project designs and simulates a 12 V to 5 V synchronous-style buck converter followed by a 3.3 V LDO post-regulator. The goals are to (a) size the power stage from first principles, (b) compare switching and linear regulation in terms of efficiency, and (c) show how an LDO can clean up switching ripple for noise-sensitive loads such as ADCs and RF circuits.

2. Specifications

Parameter	Value
Input voltage	12 V
Output voltage	5 V (buck), 3.3 V (after LDO)
Load current	1 A maximum (load step 0.2 A to 1 A)
Switching frequency	500 kHz
Inductor ripple target	about 30% of load current
Output ripple target	below 50 mV pk-pk

3. Design Calculations

Duty cycle: $D = V_{out} / V_{in} = 5 / 12 = 0.417$

Inductor: $L = V_{out}(1 - D) / (f_{sw} \cdot \Delta I_L) = 5 \times 0.583 / (500\ \text{kHz} \times 0.3\ \text{A}) \approx 19.4\ \mu\text{H}$. The next standard value, **22 µH**, was chosen, giving an actual ripple of $\Delta I_L = V_{out}(1 - D) / (f_{sw} \cdot L) = 0.265$ A.

Output capacitor: The minimum capacitance from $C \geq \Delta I_L / (8 \cdot f_{sw} \cdot \Delta V)$ is about 1.3 µF for a 50 mV ripple budget. A **22 µF** ceramic was chosen because ceramic capacitance falls under DC bias and because it lowers ripple and improves load-step behaviour. Estimated ripple is about 3.0 mV (capacitive) plus 1.3 mV (5 mΩ ESR × 0.265 A), about 4.3 mV in total.

LC resonance: $f_0 = 1 / (2\pi \sqrt{(LC)}) \approx 7.2$ kHz.

Why not an LDO alone from 12 V: An LDO dropping 12 V to 5 V at 1 A has efficiency of 5 / 12 ≈ 42% and dissipates (12 − 5) × 1 A = 7 W as heat, which would need a heatsink. A buck converter loses only conduction and switching losses.

4. Circuit Architecture

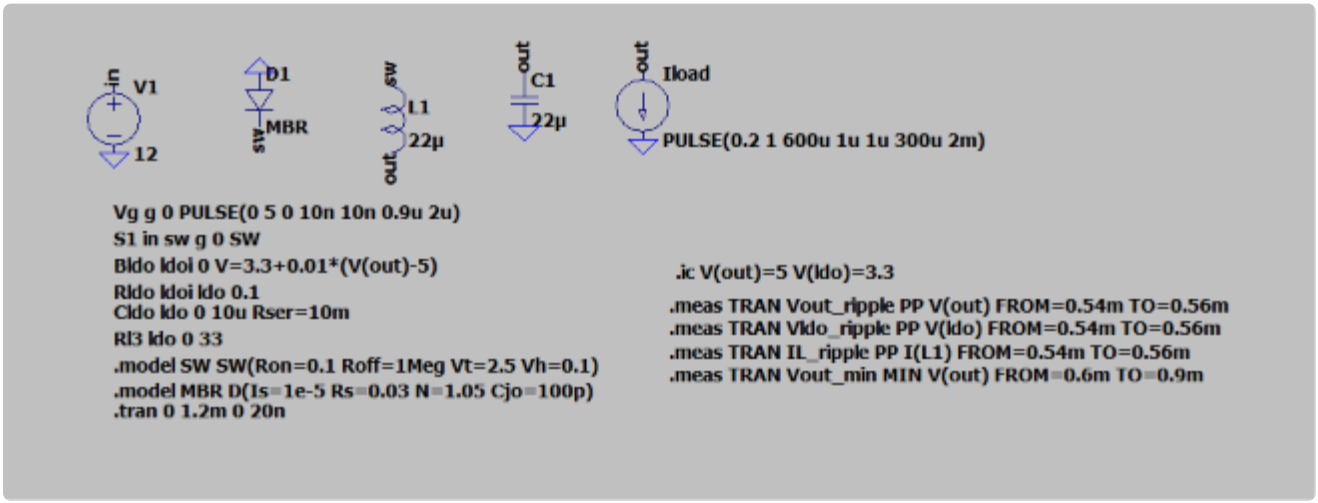


Figure 1. Buck converter with LDO post-regulator. Component values: L1 = 22 µH (DCR 30 mΩ), C1 = 22 µF (ESR 5 mΩ), S1 Ron = 0.1 Ω, D1 Schottky-type model, Cldo = 10 µF, Rl3 = 33 Ω.

The converter is simulated open loop with a fixed on-time of 0.9 µs in a 2 µs period (45% duty). This is slightly above the ideal 41.7% to compensate for conduction drops. The LDO is a behavioural model that passes 1% of the input deviation to its output (40 dB PSRR) around a 3.3 V reference.

5. LTspice Simulation Results

5.1 Startup and load step

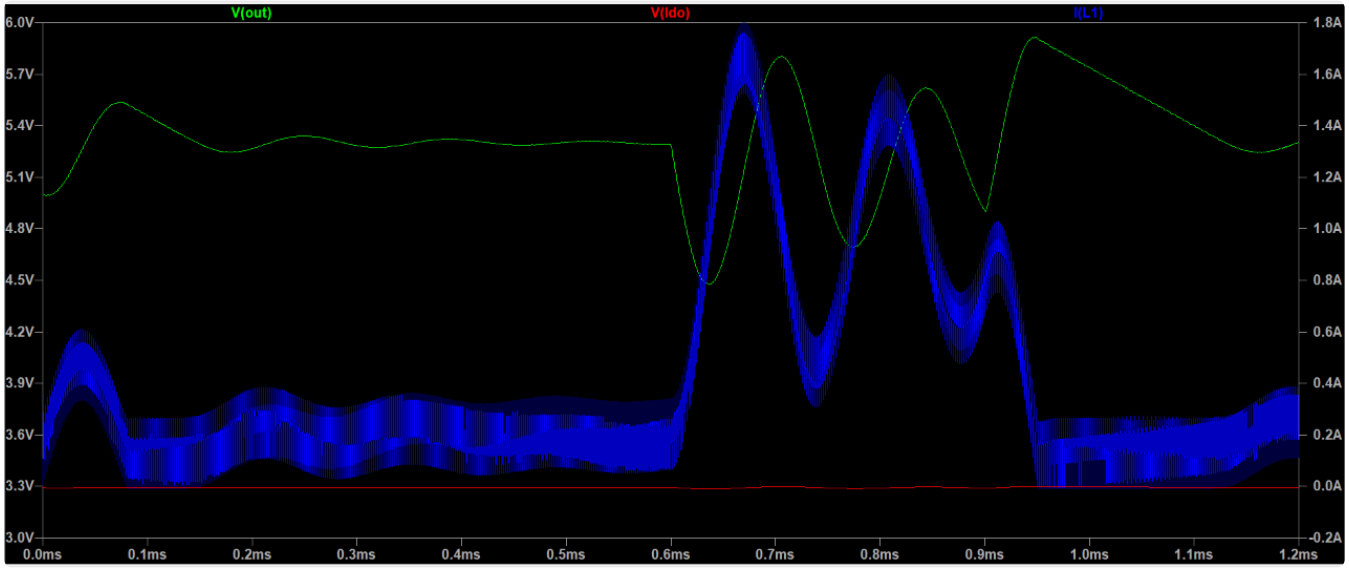


Figure 2. V(out) and I(L1) over the full run. The load steps from 0.2 A to 1 A at 0.6 ms.

Initial conditions (`.ic V(out)=5 V(ldo)=3.3`) were used to remove the open-loop startup overshoot, which otherwise reached about 10 V with inductor current peaking above 5 A because the loop has no soft-start or feedback. After startup, Vout settles near 5.3 V. The load step to 1 A makes Vout dip to 4.475 V, after which it rings at about 7 kHz, in agreement with the LC resonance in Section 3.

5.2 Steady-state ripple

Steady-state output ripple and inductor current can be observed in fig 2 from 0.54 to 0.55 ms.

5.3 Measured versus target

Metric	Target / hand calculation	Measured
Output ripple (5 V rail)	below 50 mV (about 4.3 mV estimated)	10.4 mV pk-pk
Inductor ripple current	0.265 A	0.279 A pk-pk
3.3 V rail ripple	as low as possible	0.083 mV pk-pk
Ripple rejection of LDO stage	40 dB (model)	about 42 dB
Vout minimum after 0.2 A to 1 A load step	n/a (open loop)	4.475 V
Ringing frequency after load step	7.2 kHz (LC resonance)	about 7 kHz

The measured output ripple of 10.4 mV is higher than the 4.3 mV estimate because the 20 µs measurement window also captures switching edges and a small amount of slow ringing, and because the estimate ignores diode and switch parasitics. The inductor ripple is about 5% above the hand calculation because the simulated duty cycle (45%) is above the ideal 41.7%. Ripple at the 3.3 V rail is about 125 times smaller than at the 5 V rail.

6. MATLAB Analysis

The script `buck_design.m` recomputes the design values, estimates efficiency versus load for the buck and for an LDO using a simple loss model, and plots the Bode response of the open-loop control-to-output transfer function.

6.1 Printed design values

Quantity	MATLAB output
Duty cycle D	0.417
Inductor L	22 µH
Inductor ripple current	0.265 A
LC resonant frequency	7.2 kHz

6.2 Efficiency versus load

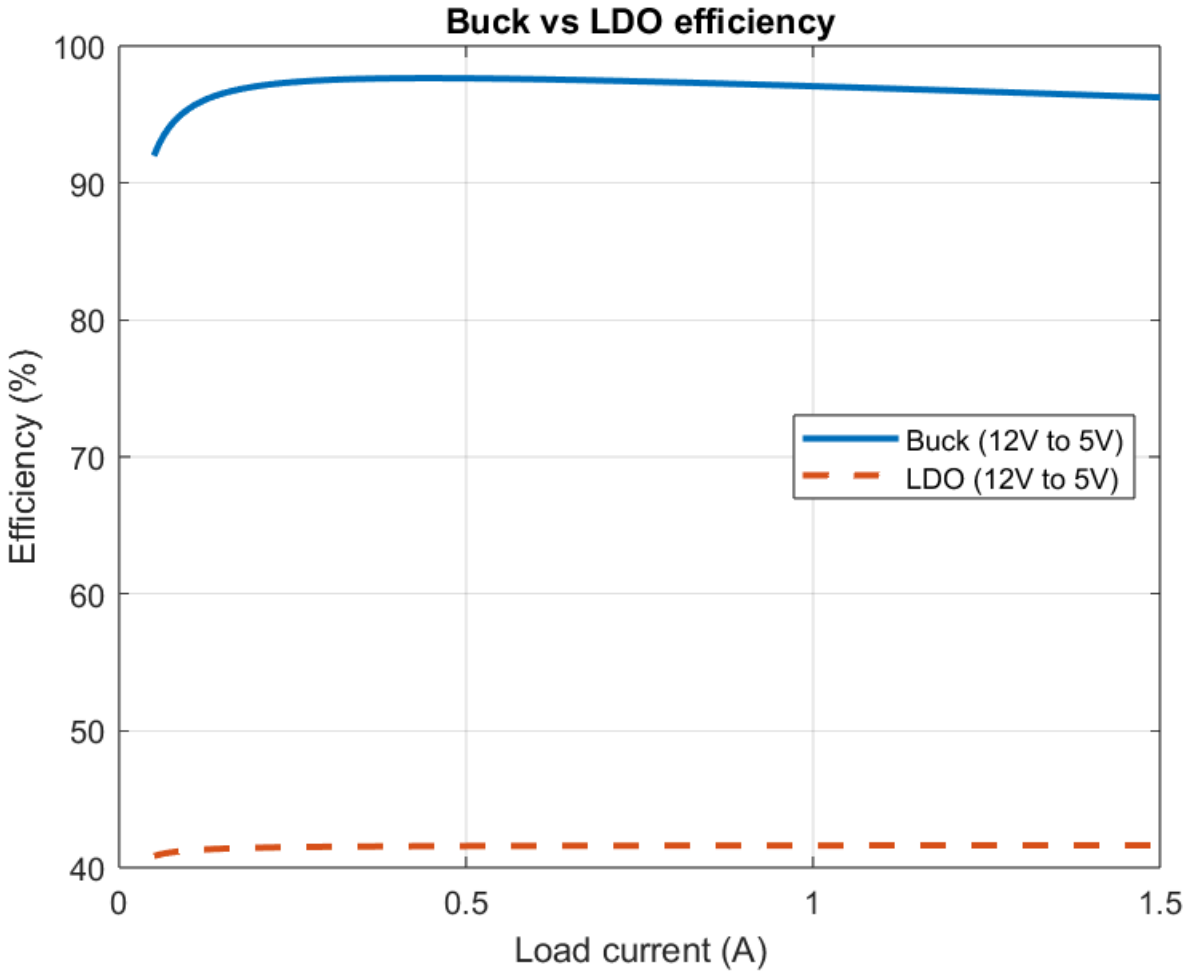


Figure 5. Estimated efficiency of the buck converter versus an LDO, 12 V to 5 V.

Load current	Buck efficiency	LDO efficiency
0.1 A	95.5%	41.3%
0.5 A	97.6%	41.6%

Load current	Buck efficiency	LDO efficiency
1.0 A	97.1%	41.6%

Power dissipated at 1 A: LDO = 6.99 W; buck = 0.15 W.

Comment: The LDO stays near 41.5% at every load, close to Vout/Vin = 41.7%, and dissipates 6.99 W at 1 A, while the buck reaches 95.5% to 97.6% and dissipates only 0.15 W, about 47 times less heat. The buck loss model includes only conduction, switching-edge and fixed losses. It ignores diode, core and AC losses, so the buck efficiency shown is an optimistic estimate rather than a measured value.

6.3 Open-loop frequency response

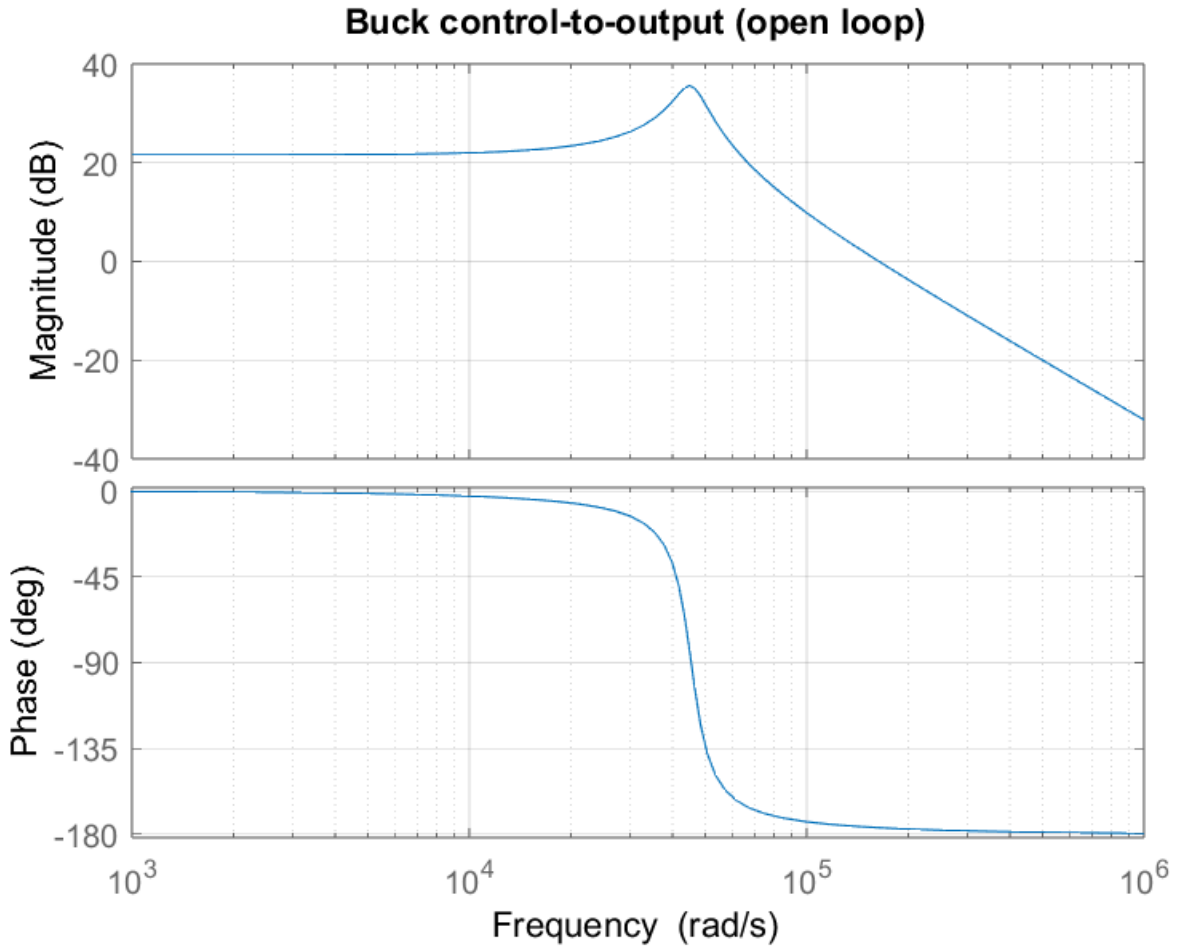


Figure 6. Control-to-output Bode plot (open loop, no damping).

Quantity	Value
Resonant peak frequency	7.16 kHz
Peak magnitude	35.6 dB

Quantity	Value
Phase change through resonance	180 degrees

Comment: The 7.16 kHz peak matches the calculated 7.2 kHz LC resonance and the roughly 7 kHz ringing in the LTspice load step. The 35.6 dB peak corresponds to Q = R·√(C/L) = 5, giving a gain of Vin × Q = 60, so the undamped response needs compensation before the loop is closed.

7. Discussion

- **Efficiency:** At a 7 V drop, a linear regulator wastes most of its input power as heat, so a switching stage is the right choice for the main step-down.
- **Noise:** Switching ripple of about 10 mV on the 5 V rail was reduced to below 0.1 mV by the LDO stage, which supports using a buck followed by an LDO for ADC and RF supplies.
- **Open-loop limitations:** Output voltage drifted to 5.3 V, the load step caused a dip to 4.475 V, and the LC filter rang at about 7 kHz. All three are symptoms of having no feedback loop.

8. Limitations

1. The LDO is a behavioural model with a fixed 40 dB PSRR. Real LDOs lose PSRR at high frequency, so rejection at 500 kHz would be lower than shown. A library part such as an LT1763 or LT3042 should be used for a realistic comparison.
2. The buck is open loop with an ideal switch and a simplified diode model.
3. The MATLAB loss model is simplified and is not a substitute for measured efficiency.

9. Conclusion

A 12 V to 5 V, 1 A, 500 kHz buck converter was designed by hand calculation and verified in LTspice and MATLAB. The simulated ripple (10.4 mV) is well within the 50 mV target, and a downstream 3.3 V LDO stage reduced ripple by about 42 dB. The work shows the key trade-off in power-management architecture: switching converters give efficiency and linear regulators give low noise, so combining them gives both.

Appendix: Files

`buck_ldo.asc` (LTspice schematic), `buck_design.m` (MATLAB script), `images/` (figures), this report.